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Design and modeling of linearly-constrained compliant spines for human-scale locomotion on rocky surfaces

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Abstract

We present a new spine solution for the locomotion of human-scale robots on steep, rocky surfaces, known as linearly-constrained spines. The spine stiffness is low in the normal direction but high with respect to lateral and bending loads. The solution differs from previous spine arrays used for small robots in having a much higher spine density and less spine scraping over asperities. We present theoretical and empirical results to demonstrate that this solution is capable of shear stresses of over 200 kPa, enabling human-scale robots to apply forces parallel to steep rock surfaces for climbing, bracing, etc. The analysis includes the effects of spine geometry, stiffness, backlash and three-dimensional loading angle to predict the overall forces possible in three dimensions of both single and opposed configurations of spine arrays. Demonstrated applications include a gripper for a "smart staff" aimed at helping humanoid robots to negotiate steep terrain and a palm that provides over 700 N in shear for the RoboSimian quadruped.

Keywords

Spines, climbing, legged locomotion, mechanism design, field robots

1. Introduction

A continuing challenge faced by human-scale robots for search and rescue, planetary exploration and related applications is to negotiate rubble and steep, rocky terrain (Chung and Khatib, 2015; Hauser et al., 2008, 2005; Kajita and Tani, 1991; Khatib and Chung, 2014). For bipedal robots, one possible solution is to equip them with a collapsible, instrumented "smart staff" in each hand (Khatib and Chung, 2014; Wang et al., 2015) to extend the reach of an extra point contact. This solution draws inspiration from hiking poles used by humans to improve stability and reduce strain (Jacobson et al., 2000). With this solution, it is further desirable if the robot can brace itself by "grasping" steep or even vertical surfaces (Figure 1). In the case of large, multi-legged robots, it is similarly desirable if the hands or feet can apply large tangential as well as normal forces on steep surfaces (Satzinger et al., 2015; Wang et al., 2016).

In both cases, various climbing technologies, some adapted from human climbing equipment and others developed for climbing robots, can enhance the ability to apply large forces. For example, soft rubber can produce a large coefficient of friction (Persson et al., 2004), helping humans or robots to climb. In challenging environments, human climbers use various gripping techniques to exploit terrain features (Amca et al., 2012; Low, 2005; Morenas Martín et al., 2013; Silva et al., 2008). Finally, hooks and surface breaking tools (Luebben, 2004) can anchor on surfaces, however these are typically slow and can damage or deface the surface.

In addition to adapting human climbing strategies, robots that negotiate rocky surfaces can use adhesives, aerodynamic effects and arrays of compliant spines or claws. Among these options, spines and claws have the advantages that they work well on rough outdoor surfaces, are unaffected by dirt or moisture, and do not require continuous power consumption.

Arrays of claws and spines have been applied to small climbing robots on rough vertical surfaces (Asbeck et al., 2006; Lee and Fearing, 2015; Liu et al., 2015; Lynch et al., 2012; Parness et al., 2015; Sintov et al., 2011; Spenko et al., 2008). In a heavier application, an octagonal array of compliant spines was used on the 8 kg LEMUR IIb planetary robot for climbing rock walls and anchoring on rocky surfaces in space (Parness, 2011; Parness and Frost, 2012; Parness et al., 2012, 2013). A large "paddle" of compliantly-supported spines was able to support an 84 kg human on

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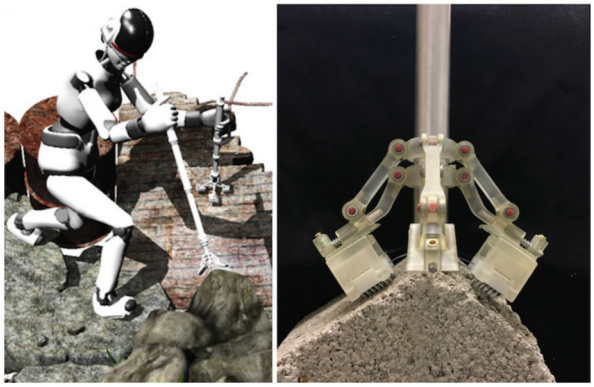


Fig. 1. Left: a humanoid using "smart staffs" negotiates rocky terrain. Right: smart staff tip is a gripper with arrays of linearly constrained spines to grasp rough surfaces.

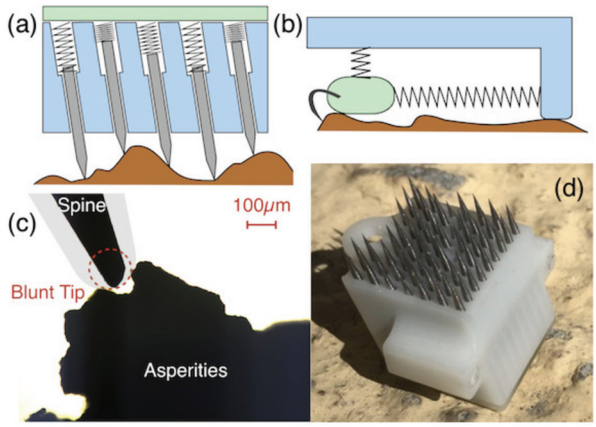


Fig. 2. Comparison between linearly-constrained spine mechanism (a,d) and compliant long-flexure spine mechanism (b). Microscopic view (c) of a new and worn spine (50 μm tip radius) engaging an asperity. Gray shaded region shows that the asperity is not usable for a spine with larger tip radius (100 μm).

a stucco wall (Asbeck and Cutkosky, 2012), but its bulky design makes it impractical for mounting on the hands or feet of a human-scale robot.

This paper contributes a new linearly-constrained spine technology that achieves a much higher spine density and shear stress than previous compliant spine designs. It also provides greater spine travel in the direction normal to the surface; hence it can be used on rough surfaces. It is suitable for mounting on the extremities of a human-scale robot that must negotiate steep, rocky terrain. Preliminary results for a palm with linearly constrained spines were reported in Wang et al. (2016). This paper presents a more detailed model of spine/surface interaction on curved as well as flat surfaces and extends the results to smart staffs suitable for bipedal humanoid robots and end-effectors for human-scale legged robots.

Section 2 presents the new spine design and modeling theory. Section 3 extends the model to arrays of tiles, with experimental validation. Section 4 presents opposed spine arrays and their corresponding limit surface in force space, with a discussion of the main design parameters. Designs based on the new spine arrays and their performance are presented in Section 5 for two different human-scale robot applications. Section 6 presents conclusions and future work.

2. Linearly-constrained spines

2.1. Spine mechanism

Inspired by insect legs (Dai et al., 2002), spines with 10-15μm tip radii catch on asperities (small bumps or pits on a surface) if the loading direction is within the friction cone of the (microscopic) local surface (Asbeck et al., 2006). When loaded in shear, they can apply modest negative (i.e. adhesive) as well as large positive normal forces.

Figure 2 (b) shows an idealization of a compliant spine mechanism that has been used on several climbing and perching robots (Asbeck and Cutkosky, 2012; Asbeck et al., 2006; Desbiens et al., 2011; Jiang et al., 2014; Liu et al., 2015; Parness et al., 2015; Spenko et al., 2008). The mechanism has a long tangential flexural element and a soft normal flexure. The tangential flexure allows spines that are tied to a common base to travel over a surface and find usable asperities without interfering with each other. This load sharing property is critical for combining arrays of small spines to generate larger forces. The soft normal flexure keeps the spine in contact as it travels tangentially to avoid missing asperities.

The compliantly-supported spine design is lightweight, has good impact resistance and is robust. However, it has some drawbacks when applied to human-scale rock climbing. The tangential flexure must be able to extend enough to guarantee sufficient travel of spines for good load sharing. This constraint limits the spatial density of spines and can lead to bulky designs for large loads (Asbeck and Cutkosky, 2012). The range of spine motion in the normal direction is also confined, which makes it hard to adapt to large scale asperities and irregular surfaces. Hence, the compliant spine mechanisms work best on surfaces like flat concrete and stucco walls. Moreover, the compliance between the base of the spines and the asperities can be undesirable for locomotion of large robots where stiff attachment benefits control.

To address the aforementioned problems, we present a new spine design, known as linearly-constrained spines (Figure 2(a) and (d)). The spine is completely straight with a sharpened tip, which initially has a tip radius of $\approx$10 μm and becomes as large as 50 μm radius after extended use. Its body is a slider that travels in a tubular cavity, angled with respect to the surface normal. A soft spring element located within the sliding channel helps to press the spine against the contact surface. This design is approximately equivalent

to a compliantly-supported spine with almost rigid tangential flexures and no potential for bending rotation. However, it is much more compact.

For a group of such spines, because of the high tangential stiffness, each spine can travel a negligible distance along the surface. Accordingly, spines must rely on whatever asperities they encounter immediately. Therefore, the percentage of engaged spines is lower than for the compliantly-supported design with elastic deflection in the tangential direction. However, as will be shown later in the paper, this limitation is compensated by a much higher spine density. Load sharing among those engaged spines is approximately the same as in the compliantly-supported design because the spines are not entirely rigid. The new design also has a large range of travel normal to the surface, which is desirable for irregular surfaces.

2.2. *Spine model and design parameters*

The performance of a spine mechanism depends on how well it conforms to a surface and engages asperities (Asbeck et al., 2006). For the linearly-constrained spine mechanism, performance depends primarily on the dimensions of the spines, their inclination angle, and backlash and friction between the spines and their channels. The main constraints and corresponding design implications are summarized below. Supporting definitions and derivations are provided in the Appendix A.

2.2.1. Spine strength. On strong asperities, the adhesion of a spine is limited by the spine's own strength which depends on its diameter d_s, the tip included angle, α, and how far it extends from its channel, l_e. When the loading location is close to the sharpened tip, as in Figure 3 (left), failure is related to the shear stress: $4F_r/(\pi d(y)^2)$ where $d(y)$ is the spine cross-section diameter at loading position y. In practice, sharp ($\approx 10\mu$m tip radius) spines quickly become blunted by tip forces on the order of tens of Newtons. A maximum expected tip force of 70 N corresponds to a minimum diameter of the tapered spine tip of $\approx$ 0.1 mm on the current design, which empirically is the steady state condition after extended use.

When F_r is applied at a worn tip, the next likely source of failure is bending, as in Figure 3 (right). There are two local maxima: the spine may yield at the upper contact, where it exits from the spine channel, or along the tapered section highlighted in red. The derivation is provided in Appendix B. The extended length, l_e, and spine diameter, d_s, are limiting factors for the upper maximum. For example, a 1 mm diameter steel spine extended out by 5 mm can withstand up to 70 N force at the tip without breaking. Under large loading forces, the channel wall corner where the spine exits may creep to form area contact instead of edge contact. This does not noticeably change the spine bending, but it can produce slightly more backlash (within 0.2 mm observed on a worn tile) which slightly affects load sharing as discussed in Section 2.2.4.

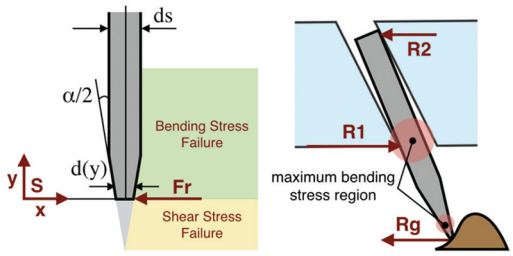


Fig. 3. Diagrams for spine strength analysis. In the left diagram, the spine is drawn with the light gray shaded region being the original sharpened spine shape, and with the solid outline of the spine corresponding to the shape after it gets worn down due to the shear stress failure. Failure is dominated by shear stress when the loading location is near the tip and by bending stress otherwise. There are two local maximums highlighted in red: the spine may yield at the upper contact, where it exits from the spine channel; or along the tapered section near the spine tip. R_1, R_2 and R_g are reaction forces on the spine.

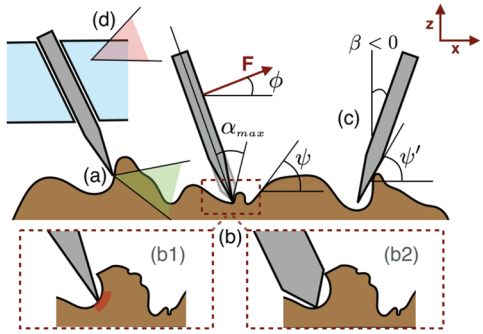


Fig. 4. Different limiting factors on spine adhesion under certain loading angle ϕ and spine inclination angle β: (a) range of spine approach angles (green shade) to prevent spine-asperity slip; (b) maximum spine tip included angle, α_{max}, to allow tip contact on small asperities (red shading highlights contact stress region). Magnifications (b1) and (b2) show the effect of tip included angle; (c) undesirable spine inclination angle causes asperity tip contact and results in low asperity strength; (d) range of loading angle (red shade) on the substrate which does not cause the spine to slip with respect to the sliding channel (spine-channel slip).

For the second local maximum, which is usually the tighter constraint, the spine tip included angle, α, becomes the limiting factor. It needs to be at least 14° for a worn spine to withstand up to 40 N force (stronger than most asperities on concrete surfaces (Wang et al., 2016)) without plastic bending near the tip.

2.2.2. Spine-asperity interaction. Another factor that affects adhesion is how the spine interacts with an asperity. Depending on how a spine is inclined and loaded with respect to the asperity, the spine-asperity interaction falls into three different cases illustrated in Figure 4: (a) the

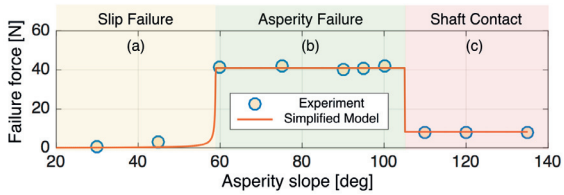


Fig. 5. Maximum shear loading force of an enlarged spine ($\alpha = 15^\circ$, $\beta = 15^\circ$, tip radius = 2 mm, $\phi = 0$, $\mu = 0.4$) on large plaster asperities with different slopes, ψ. The behavior falls into three regimes, matching cases (a), (b) and (c) in Figure 4. The model is based on equation (1) and (2).

spine slips along the surface when the tip contact force is outside the local friction cone; (b) the spine tip is engaged with the surface (tip contact) and (c) the spine tip cannot reach the asperity due to shaft side contact when the spine inclines too much.

For this analysis, the failure force F due to friction (a) is

$$F = \frac{\mu \cos\psi}{\cos\psi - \mu \sin\psi} F_P \tag{1}$$

where F_P is the compression force provided by the soft spring in the sliding channel, and ψ is slope of the contact surface (asperity slope). F rises rapidly as the force vector approaches the friction cone and can be approximated as a step function when F_P is small. The failure force is then limited by the asperity strength under tip contact (b) which is modeled as a constant (further discussion in Section 2.4).

As seen in the enlarged inset Figure 4 (b2), if the spine tip included angle, α, is too large, the spine may fail to engage an asperity at its tip. Similarly, in Figure 4 (c), if the spine inclination angle, β, is too small, the spine may engage the asperity in side contact. In both cases, the strength of the asperity is significantly reduced. Models of the strength of cementitious materials like stucco and cement are available in the literature (Momber, 2004). However, the materials are brittle and prediction of asperity strength is difficult. For our purposes, it suffices to note that asperity strength grows with cross-sectional area, which depends on the roughness of the surface. We lump this effect into a constant unknown coefficient $C_i(\psi, \beta)$ that will be empirically fit in Section 3.2

$$C_i(\psi, \beta) = \begin{cases} c_i & \text{if } \psi \le \beta + \pi/2 - \alpha/2 \\ 1 & \text{else} \end{cases} \tag{2}$$

The contact surface slope ψ is the slope of the spine tip rather than asperity slope in the case of shaft contact

$$\psi'(\psi, \beta) = \begin{cases} \psi & \text{if } \psi \le \beta + \pi/2 - \alpha/2 \\ \beta + \pi/2 - \alpha/2 & \text{else} \end{cases} \tag{3}$$

To evaluate the effects of varying spine load angle, we constructed magnified asperities of plaster and pulled against them in pure shear with a magnified spine with a 2 mm tip radius. Figure 5 shows the results for a range of asperity slope angles, ψ (5 trials for each angle), which match the cases depicted in Figure 4. In regime (a), the performance is limited by surface friction (equation (1), $\mu = 0.4$); in regime (b) and (c), we assume constant asperity strengths that are empirically measured, and slopes represented by equation (3).

2.2.3. Spine-channel Interaction. There are two additional constraints, which concern the sliding of spines within their channels. Friction μ_s between the spine and its channel should prevent the spine from sliding when the spine tile is loaded. Figure 4 shows that for a range of loading directions (a) the associated friction cone for spine/channel sliding is (d). To account for this limit we can introduce a coefficient, $C_c(\phi, \beta)$, which becomes 1 when no slip occurs and otherwise is limited by μ_s and β (details are given in Appendix C)

$$C_c(\phi, \beta) = \sin(\beta + \pi/2 - \phi - \arctan\mu_s)(1 + \mu_s^2) \tag{4}$$

The second constraint is due to particles that become dislodged from the surface and can jam the spines. For a given tip included angle α, channel clearance, and friction coefficient, particles of a certain dimension, ϵ, may either get stuck, or will be pushed out by the spine as it extends. Using the derivation in Appendix C, we find that the coefficient of friction, μ_2, between the channel and dirt particle should ideally be lower than the coefficient of friction, μ_1, between the spine and the particle. Empirically, we found that a prototype with channels made from 7075 T-651 aluminum encountered problems with particle jamming ($\mu_2 = 1.35$) whereas a prototype made from polypropylene, with coefficient of friction $\mu_1 \approx 1$, did not.

2.2.4. Spine load sharing and conformability. Each spine supports a relatively small force and therefore the spine mechanism needs to share its load among multiple spines. For the linearly constrained spine design, load sharing is primarily a function of spine contact stiffness and backlash. Spine contact stiffness is the equivalent spring stiffness between the spine mechanism and the tip contact due to spine deflection, projected along the loading direction. For a given set of materials, it depends on spine shaft diameter, d_s, and the extended length, l_e. The contact stiffness of a single spine was characterized to be 17.5 N/mm in Wang et al. (2016) with compliance mainly from the slim tapered tip. Note that a less stiff spine will in general have a smaller load than its neighbors; this effect protects those spines that have extended farthest, with the largest value of l_e.

Backlash, s, is the maximum tip travel the spine can have along the surface under negligible loading force. It is a function of the nominal spine/channel clearance and of manufacturing tolerances. If the spine total length is l and

length of the portion extended out of the channel is l_e, the backlash can be calculated with

$$s = \frac{l}{l - l_e}\epsilon \tag{5}$$

where ϵ is the clearance between the spine and the channel. Note that the change on spine inclination angle due to backlash is within 2deg in most cases, thus is not considered in the model. Details are provided in Appendix A. On fine surfaces, with small asperities, backlash can significantly reduce load sharing, as discussed in Section 3.

Another important specification is conformability, which reflects the ability a group of spines to match the contour of a surface. Conformability increases with maximum extension, l_e. The spring force, k_n, should be low so that one does not need a significant force to press a group of spines against a surface, but it must also be sufficient to extend each spine.

2.3. Spine design guidelines

The constraints on spine strength, asperity loading and friction suggest some design guidelines. Based on the surface rugosity, we choose the desired maximum extension, l_{e_max}. Then the shaft diameter, d_s, and tip included angle, α, are chosen to make the spine as sharp as possible, without premature breaking or bending and without tending to jam dirt particles between the spine and channel.

The spine inclination angle, β, is then set to be large enough to reach deep asperities but not so large that undesirable side contacts become frequent. This condition depends on how sharp the asperities are. Knowing the RMS roughness of the surface does not suffice; one should also know the mean peak height and, ideally, the kurtosis (Asbeck et al., 2006). In practice, an angle of $\beta \approx 15^\circ$ works well on many concrete, stone and stucco surfaces.

Materials choices are also important: the spines should be as hard as possible without becoming brittle and the channel should be hard, to minimize deflections, and have a low coefficient of friction to avoid jamming particles.

With the aforementioned considerations, the following parameter values were chosen for spines and spine arrays tested in this paper: $l_{e_max} = 5\,\text{mm}$, $d_s = 1\,\text{mm}$, $\alpha = 15^\circ$, $\beta = 15^\circ$, $k_n = 5\,\text{mN/mm}$. The spine material is high speed steel (HRC63) and the channel wall material is a polypropylene/ABS blend (Protolab Xtreme White 200).

2.4. Modeling the surface

To progress from a single spine to an array of spines it is necessary to adopt a model that accounts for the spatial distribution of asperities and the stochastic nature of spine/asperity contact strength. In contrast to previous compliant spine mechanisms, which allowed each spine to travel several millimeters before encountering an asperity, the spines in a linearly-constrained array will have very little travel along the surface before the array comes to rest.

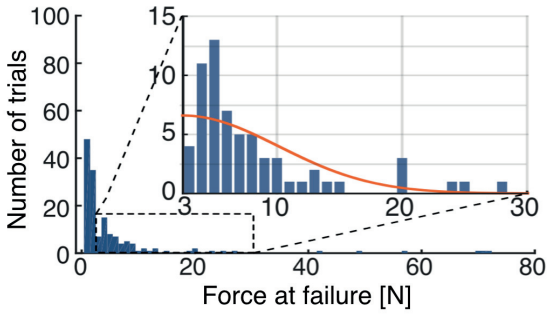


Fig. 6. Empirical data from 150 measurements with a single spine on fine concrete for estimating relevant probability distributions. Magnified area shows the portion due to asperity failure, which is fitted with a truncated Gaussian.

Therefore, instead of a Poisson spatial distribution as in Asbeck et al. (2006), it suffices to take the total area of asperities divided by the total surface area to describe asperity density. On some asperity-abundant surfaces such as the two concrete surfaces tested in Wang et al. (2016), the ratio can be up to 34% and 42%.

However, not all asperities are equally useful. As seen in the previous section, the force available from each spine is a function of several factors including the local surface angle, the asperity strength, and the stiffness and strength of the spine itself. Let M represent the relevant failure mode and let Φ_M represent the maximum force that corresponds to that failure mode. The force contributed by each spine in an array is then a function of its stiffness and deflection, subject to the maximum force limitation. We start by writing the travel of the spine array in terms of stiffness, k, and backlash, S (upper case for a random variable), which we assume to be uniformly distributed because not all spines will be at their maximum backlash. The travel (less than 1mm in most cases) is limited by the maximum force on the "worst case" spine that is at its limit

$$X = \min_{i=1}^{n}(\,\Phi_{Mi}/k + S_i\,)\,. \tag{6}$$

The force each spine contributes is then limited by the spine array travel

$$F_i = k\max\{X - S_i, 0\}. \tag{7}$$

Although the various factors affecting the maximum spine/asperity contact strength have been discussed in the previous section, the practical results for spines on real surfaces are stochastic. We therefore fit a probability density function to empirical results to obtain Φ_M. For example, Figure 6 shows typical data for spines loaded parallel to the surface ($\phi = 0$) on concrete blocks. Additional examples are provided in Wang et al. (2016).

Extending the results to a collection of spines we can write

$$E[F] = \sum_{i=1}^{n}\iiint F_i f_{\Phi_{Mi}} f_{Si} f_{M_i} dM_i dS_i d\Phi_{Mi} \tag{8}$$

where $E[F]$ (written simply as F hereafter) is the expectation of the spine tile adhesion, f is the probability density function of the corresponding random variable, and random variables M_i, Φ and S_i indicate the failure mode, the maximum adhesion a spine can contribute under the corresponding failure mode, and the backlash for an engaged spine, respectively. This model does not have a closed-form solution but, using distributions such as those in Figure 6 and measured values for k and S, we can predict the effect of increasing numbers of spines in an array with Monte Carlo simulation. Detailed discussion of the selected probability distributions is in Wang et al. (2016). As an example, Figure 7 shows the predicted results for the case $\phi = 0$ (pure shear loading) on smooth and rough concrete, compared to empirical results for 20 trials. Forces were measured with an ATI™ Gamma load cell with a resolution of ≈ 0.05 N.

Figure 7 shows that arrays of spines achieve larger adhesion on coarse concrete surfaces but with a greater standard deviation because the asperities are more stochastically distributed. Scaling efficiency in all cases decreases with the number of spines but can be significantly improved by reducing the spine backlash. This is because with larger backlash the spines in the same tile tend to have more widely distributed individual backlash, which makes a larger portion of the spines fail to assume a load. Tiles with fewer spines tend to have longer free travel before any spine overcomes its backlash, and thus the actual mean backlash is smaller, which explains the nonlinear scaling.

3. Spine tile limit surface

3.1. Spine array adhesion with varying loading angle

The data in Figures 6 and 7 are for a single loading angle. It is impractical to build up an empirical statistical model for every loading angle. However, we can predict the effect of variation of loading angle if we have a model of the asperity slope distribution.

As Figure 5 shows, for a given loading angle, ϕ, the available force for low asperity slopes is initially very small, limited by slipping, and then nearly constant over a range of useful angles in regime (b). The effect of increasing ϕ is effectively to curtail the range of useful asperity slopes. Thus, if we increase ϕ from 0 to 10°, we reduce the usable range of asperity slopes from $60^\circ < \psi < 100^\circ$ to $70^\circ < \psi < 100^\circ$. Doing so also reduces the density of usable asperities on a surface, and this effect is what practically limits the maximum loading angle. In Asbeck et al. (2006) it was determined that the density of usable asperities follows an exponentially decaying trend with an increase in the desired asperity slope. For the region of interest ($\psi > 60^\circ$) this trend is nearly linear, therefore its probability density function, f_Ψ, can be simplified knowing the range of slopes for usable asperities, $\delta\psi = \psi_{max} - \psi_{min}$, and considering that the integral of the probability density function should be unity

$$f_\Psi(\psi) = \frac{2}{\delta\psi} - \frac{2}{\delta\psi^2}(\psi - \psi_{min}). \tag{9}$$

As discussed in Section 2.2.3, there are two other factors that are also associated with loading angle, which are the effects of spine shaft contact, modeled with a coefficient $C_i(\psi, \beta)$ (case (c) in Figure 4, equation (2)), and slippage of the spines in their channels, modeled with another coefficient $C_c(\phi, \beta)$ (equation (4)). The parentheses denote that these effects are subject to conditions on the associated variables (the constant becomes 1 if the condition is not satisfied).

In summary, the overall adhesion at any loading angle can be computed with the integral over all the usable asperity slopes considering both spine shaft contact and channel slip effects

$$F(\phi) = \frac{\int_{\psi_{min}(\phi)}^{\psi_{max}(\phi)} C_i(\psi, \beta)\, C_c(\phi, \beta) f_\Psi(\psi)\, d\psi}{\int_{\psi_{min}(0)}^{\psi_{max}(0)} C_i(\psi, \beta)\, C_c(0, \beta) f_\Psi(\psi)\, d\psi} F(0) \tag{10}$$

where $\phi = 0$ is the shear direction.

The usable range of slopes $[\psi_{min}\ \psi_{max}]$ can be estimated based on the surface coefficient of friction μ and the maximum loading angle, ϕ_{max}, occurring in the experiment

$$\begin{cases} \psi_{min}(\phi) = \phi - \cot^{-1}\mu \\ \psi_{max}(\phi) = \phi_{max} - \cot^{-1}\mu \end{cases} \tag{11}$$

Combining equations, we estimate the spine array adhesion at any loading angle with the following procedure.

1. Use the single-direction adhesion model, equation (8), to compute F(0) with single-spine statistics characterized along the shear direction.
2. Gather single-spine data at another loading angle, ϕ_1, to compute $F(\phi_1)$ using the same empirical method.
3. Calculate the asperity slope range $[\psi_{min}\ \psi_{max}]$ with measured surface coefficient of friction and maximum loading angle, using equation (11).
4. Solve for constant c_i in equation (2) using $F(0)$ and $F(\phi_1)$ which have been computed (equation (10)) based on two measured distributions at two different angles.
5. Having determined the unknowns in equation (10), we use it again to compute the adhesion limit, $F(\phi)$, at any angle.

3.2. 3D limit surface

If we let the original, preferred loading direction be in the (x, z) plane, with z perpendicular to the surface, we now consider the effect of pulling in the lateral direction, y. The load is now described by two angles, (ϕ, θ) measured with respect to the x and y axes. Assuming that the surface

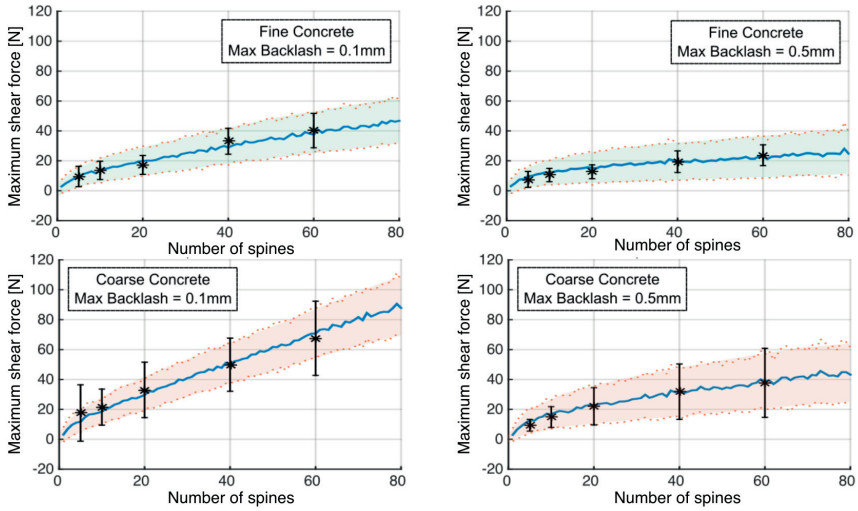


Fig. 7. Tile adhesion with different numbers of spines and backlash conditions on different surfaces. Asterisks and error bars are the mean and standard deviations of 20 experimental trials. The solid line and shaded area are the mean and standard deviation, respectively, of simulation results.

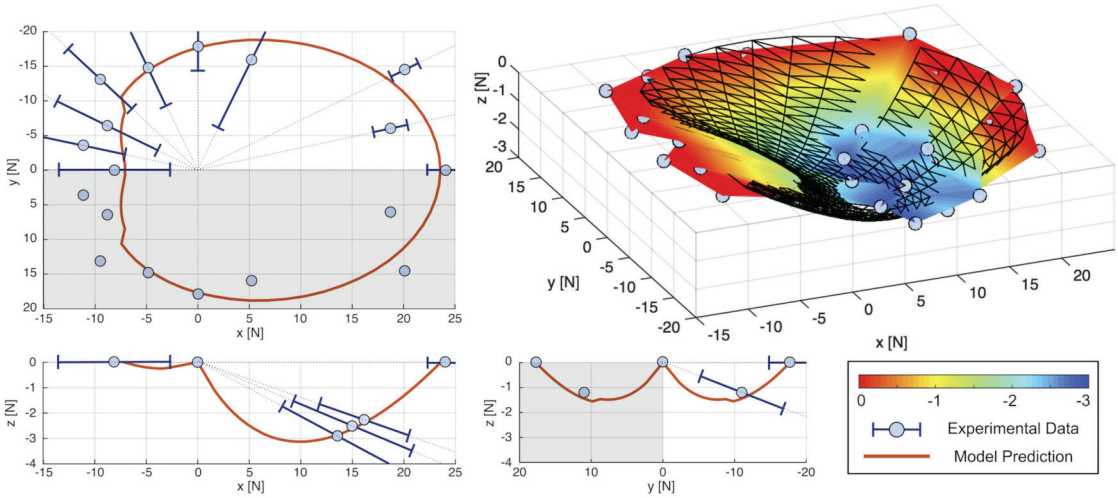


Fig. 8. Model prediction and experimental data of a single tile 3D limit surface. Each point (mean and standard deviation) is computed from four or more points in a region (300 measurements in total). Surface is symmetric about the y axis; gray shaded region is mirrored. Black mesh in perspective view represents predicted surface.

is isotropic, we define an equivalent inclination angle, β', when pulling in a direction defined by ϕ and θ

$$\beta'(\theta) = \arcsin(\sin\beta\cos\theta) \tag{12}$$

We can replace β with β' in equation (3) to compute the 3D limit surface.

3.3. *Experiments and results*

For comparison with the computed 3D limit surface, a concrete block was mounted atop the same ATI load cell used for the results in Figure 7 and 300 loads were applied in randomly chosen directions until failure. The corresponding values of (ϕ, θ) were recorded, along with the maximum force. The data were then binned according to their polar coordinates with a bin size of 3° in ϕ and 18° in θ to compute the mean and standard deviation. Figure 8 shows a comparison of the computed and measured results. Pulling in the (x, z) plane ($\theta = 0$) is optimal. As in the 2D case, increasing ϕ results in fewer usable asperities and smaller loads. As θ increases, the modified inclination angle, β', becomes less favorable. Ultimately, even small values of ϕ will allow spines to slip off the asperities; this effect is responsible for the non-convex region for $x < 0$

Figure 8 also shows large standard deviations, especially for large values of ϕ. This effect illustrates the stochastic nature of spine/asperity adhesion, which is due to stochasticity of the asperity strength, spine backlash and the fact

that the adhesion of a spine array is limited by the weakest spine contact rather than the average spine performance. Although the standard deviations are large, the averages are predictable and repeatable, as seen in the 2D case in Figure 7. Multiple methods can be applied to handle this randomness, such as performing a loading test before actually relying on the contact, estimating the contact quality based on a measured contact stiffness during loading (higher stiffness means more spines engaged), and using a differential load sharing mechanism to allow re-engaging of the spine tile, which will be discussed in Section 5.

3.4. Extended limit surface with friction

The previous discussion has considered load forces either parallel to or slightly away from a surface, as would occur for a robot climbing a vertical surface. However, when used at the tip of a staff as in Figure 1 there will often be some positive normal force (i.e. $\phi < 0$). In this case, there is a frictional force in addition to the contribution from the spines. The spines themselves are pressed against the surface by the springs, k_n, with a light force. Therefore, for the spines the condition is the same as when $\phi = 0$. However, the substrate of the tile array, containing the channels, presses against the rock with a given normal force and corresponding friction

$$F_e(\phi,\theta)\,cos|\phi| = F_e(\phi,\theta)\,\mu_t sin|\phi| + F(0,\theta) \qquad (13)$$

where $F_e(\phi,\theta)$ is the adhesive force magnitude for the extended limit surface, and μ_t is the coefficient of friction between the tile substrate and the surface. The small spring force is not included. The equation can be simplified as

$$F_e(\phi,\theta) = \frac{F(0,\theta)}{cos|\phi| - \mu_t sin|\phi|} \qquad (14)$$

where $cos|\phi| - \mu_t sin|\phi| > 0$ should be satisfied. In the limit, as the force magnitude becomes very large, the contact behaves as a standard frictional contact with a coefficient of friction μ_t. Conversely the effect of the spines is most noticeable at light loads. An *effective coefficient of friction* can be estimated using equation (14) by substituting F_e with the maximum possible loading force for a specific application and solving for $|cot(\phi)|$. Figure 9 shows how the effective coefficient of friction varies as a function of maximum load force under different surface conditions.

4. Opposed spines mechanism

Unidirectional spine arrays can take loads primarily in shear, with a small amount of force away from the surface. To support substantial normal forces, a solution is to use opposed arrays of spines with an internal grasp force. In this section, we consider such a mechanism and how its properties affect the aggregate limit surface for a pair of opposed spine arrays.

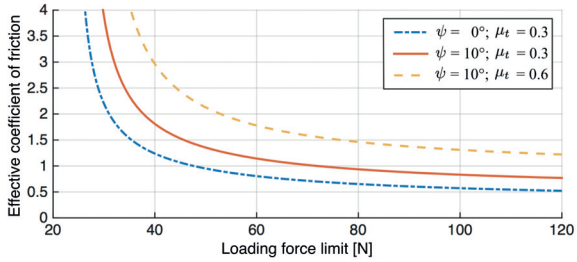


Fig. 9. Effective friction of a spine tile on different slopes and coefficients of friction between the tile substrate and surface. The effective friction decreases with increasing load force. The result is based on 60-spine tile with 0.5mm backlash on a fine concrete surface.

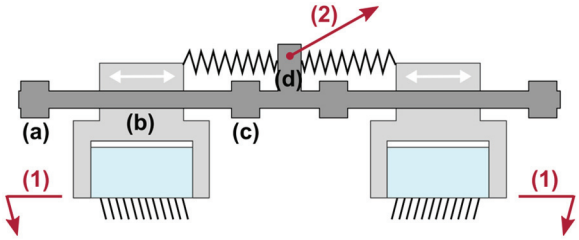


Fig. 10. Schematic of an opposed micro-spine mechanism: spine array (b) slides along the shaft, constrained by outer (a) and inner (c) hard stops. Arrows (1) show motion to engage a surface. Arrow (2) shows external load.

4.1. Mechanism design

Figure 10 is a diagram of a mechanism used for loading two opposed arrays of linearly constrained spines. Each array (b) slides on linear rails, with motion limited by external (a) and internal (c) hard stops. An external load is applied (d) to the rails and transmitted to the arrays via springs. To engage the device, we first pull the arrays apart, stretching the springs, then press the opposed spine mechanism gently against a surface and release the springs, which provide an internal force. The arrays move a small distance toward each other as the spines engage. At this point the frame is essentially rigid in the direction normal to the surface but allows some motion parallel to the surface, depending on the spring stiffness and the hard stops. To release the device, it suffices to pull the arrays slightly away from each other using, for example, a cam or toggle mechanism.

4.2. Modeling

When the center part (d) is loaded, the normal adhesion of the tiles is transmitted through the sliding frame while the shear adhesion is transmitted via the springs. Because the mechanism will be used at the end of a "smart staff" as in Figure 1, we assume that it can be kept parallel to the surface. With this kinematic constraint, the limit surface is the sum, rather than the minimum, of the contribution from each tile.[1]

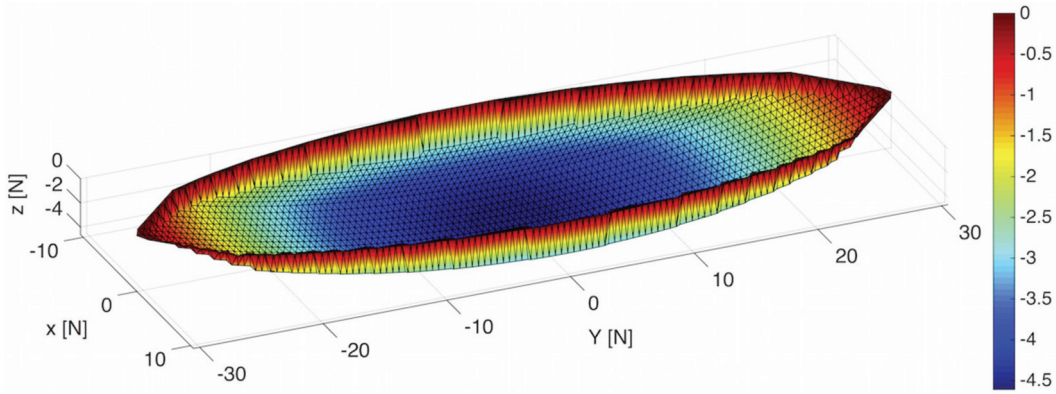


Fig. 11. 3D limit surface of the opposed mechanism on fine concrete surface (16.5 N preload, outer hard stop at 19.5 N).

As defined in Appendix D, the overall limit surface is computed by summing the contribution from each tile, taking into account the offset produced by the internal preload force and the limits imposed by the hard stops. However, if either tile completely loses its grip, the internal springs will cause the entire grip to fail.

The resulting limit surface is symmetric about the x and y axes, as seen in Figure 11. Note that the limit surface is longest in the y direction (in contrast to a single tile) because the forces from each array are added in the y direction whereas they are subtracted in the x direction. Hence the preferred loading direction is now transverse to the sliding rails.

The overall performance of the opposed-array mechanism depends on the preload and the hard stops. Increasing the internal preload increases the ability to support loads primarily in the normal direction. However, if the preload is too high, one of the tiles will quickly reach its limit. Similar considerations apply to the outer and inner hard stops. Details of the model are provided in Appendix D. When one of the tiles reaches its respective outer hard stop, additional external loads will be borne entirely by that tile. The stop should be chosen in combination with the preload and spring stiffness to maintain a desired minimum force on the opposite tile. The inner hard stop determines when one of the tiles will release, causing the entire mechanism to detach before damage occurs.

4.3. Experiments

An opposed array mechanism was tested gripping a concrete block. The internal preload was set to 16.5N, with outer hard stops at 19.5N. For each test, the arrays were pulled apart and the mechanism was loaded with a combination of shear and normal forces. As in previous tests, the concrete block was mounted to a load cell. Figure 12 shows results for 100 limiting loads in the (x, z) plane, reflected about the central axis due to the symmetry of the device. As in previous experiments, the data are binned for angular ranges of 4° and each red dot represents the average of four or more data points within a bin.

As in previous experiments, the standard deviation is roughly 30% of the adhesion, which is similar to the statistics for the single tile limit curve (Figure 8). The standard deviation when pulling primarily in the normal direction is smaller because variations in load direction have a small effect, as compared to pulling primarily in shear. Label (a) indicates when one of the tiles reaches its adhesion limit. Label (b) indicates when one of the tiles reaches its outer hard stop.

In summary, an opposed array mechanism with internal preload and sliding tiles permits loading in any direction. The preferred loading direction is now perpendicular to the rails. The internal preload and the placement of hard stops determine the limiting loads.

5. Applications

This section introduces two applications for arrays of linearly constrained spines. The first uses opposed spine arrays at the tip of a "smart staff" for a humanoid robot in rough terrain. The second uses unidirectional floating tiles, each of which contains an array of spines, intended for the foot of a large multi-legged robot.

5.1. Enhanced locomotion with SupraPed

In the SupraPed project (Wang et al., 2015), a pair of poles are provided to a humanoid robot to improve its locomotion capability as shown in Figure 1. The friction available at the tip of each pole significantly affects performance. Figure 13 shows a dynamic model of a humanoid using poles, with the maximum force along each pole limited to 20% of the robot's weight. The support region is computed using a constraint-based method, as described in Bretl and Lall (2006, 2008). Any humanoid with its center of mass (CoM) projected within the support region is considered to be stable.

With a coefficient of friction between the pole and a rock surface of 0.5, the support region is increased by a factor of 2.2 over the case with no poles. With spines, the effective

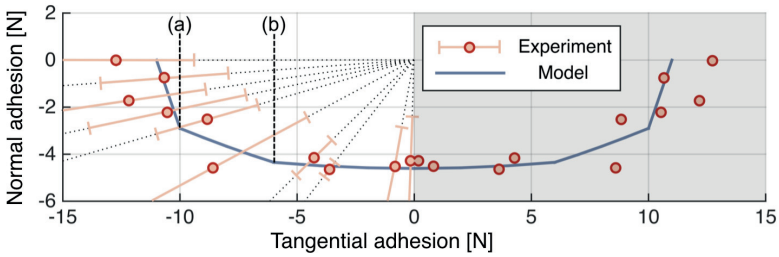


Fig. 12. Cross-section of a limit surface in (x, z) plane for gripper with opposed spines. Each data point is an average of four or more trials, with 100 loads in total. Points in gray shaded region are mirrored, assuming symmetry. Label (a) indicates the left tile reaches its adhesion limit; label (b) indicates where the left array hits the outer hard stop.



Fig. 13. Support region for a humanoid: blue circular panels are local approximations of surfaces for local motion planning; orange polygons show estimated support regions computed using constraint-based method (Bretl and Lall, 2008). Force on staff is limited to 20% of robot's weight. (left) humanoid without staffs, (middle) SupraPed with $\mu_t = 0.5$ increases support region by factor of 2.2, (right) SupraPed generates with gripper increases effective friction coefficient to 2.0 and enlarges support region by 3.4 over unassisted case.

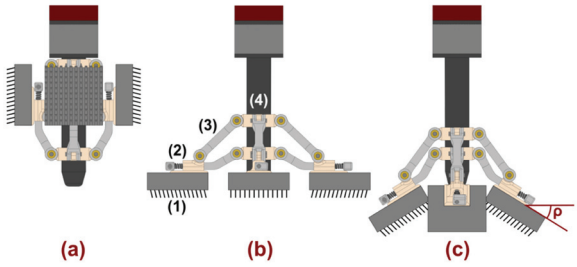


Fig. 14. Opposed spine mechanism for smart staff: (a) retracted, (b) gripping a flat surface, (c) gripping a convex surface. Elements include: (1) spine tiles; (2) preload spring and slider (prismatic joint); (3) linkages; (4) center rings (prismatic joint). ρ is spine tile inclination (same for all tiles).

coefficient of friction (Section 3.4) easily increases to 2.0, which expands the region of support by a factor of 3.4.

For gripping flat and convex surfaces, the mechanism shown in Figure 1 and Figure 14 is attached to the tip of a pole. Four tiles of linearly-constrained spines are actuated by a spring-loaded linkage. The upper ring can slide along the pole; the lower ring is fixed. Each tile of spines creates a five-bar mechanism in combination with the sliding ring. A prismatic sliding joint is located on the back surface of the tile, with a spring that pushes the tile inward (toward the pole) when it is not loaded. When not gripping, the mechanism can be retracted by actuating the sliding ring upward (Figure 14 a). To grip, the sliding ring is actuated downward to engage the tiles with a surface (b) and compress the preload spring. After the sliding ring is released, the preload springs load the tiles in the local shear direction to engage with a surface and provide adhesion, in the same manner as for the opposed spine array discussed in Section 4. The only difference is that the surface can now be either flat or convex.

The angle of inclination of the tiles is ρ (Figure 14). If we let $f_m(\phi, \theta)$ be the limit surface for a tile, the limit surface for the entire mechanism, $g_m(\phi, \theta, \rho)$, becomes

$$\begin{aligned} g_m(\phi,\theta,\rho) = {} & f_m(\phi(0,\rho),\theta(0,\rho)) \\ & + f_m\left(\phi\left(\frac{\pi}{2},\rho\right),\theta\left(\frac{\pi}{2},\rho\right)\right) \\ & + f_m(\phi(\pi,\rho),\theta(\pi,\rho)) + f_m\left(\phi\left(\frac{3\pi}{2},\rho\right),\theta\left(\frac{3\pi}{2},\rho\right)\right) \end{aligned} \tag{15}$$

where $\phi(\xi, \zeta)$ and $\theta(\xi, \zeta)$ are mapped from global ϕ and θ to a local coordinate frame that is rotated about its z axis by ξ and about its y axis by ζ. The limit surface and effective coefficient of friction can then be computed using eq. (15) and the enhanced friction method in Section 4.

Figure 15 shows the effective coefficient of friction achieved with the presented mechanism under various conditions. If the contact force on the staff tip is limited to 100 N (20% of a 50 kg humanoid), grasping a convex feature for which $\rho = 30^\circ$ results in approximately 200% increase in the effective coefficient of friction as compared to a flat surface. Similarly, the effective coefficient of friction can be improved from 0.5 to 2 with the mechanism (60-spine tiles, 0.5 mm backlash, 21 N shear adhesion) grasping a 20° convex concrete feature. Tiles with better adhesion (60-spines, 0.1 mm backlash, 40 N shear adhesion) can achieve an effective friction coefficient of 2 on flat surfaces for a contact force up to 135 N, which could greatly increase the stability of a humanoid robot.

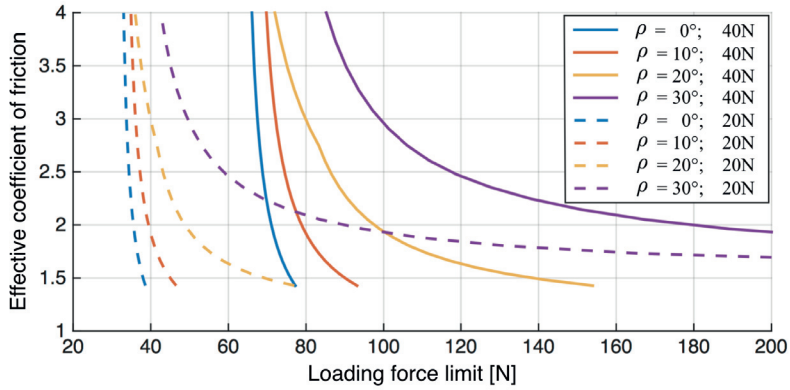


Fig. 15. Effective coefficient of friction created by the SupraPed spine mechanism for convex features with different inclinations, ρ and shear force capabilities of 20 N or 40 N. Computed results assume an array of 60 spines with 0.5 mm backlash on a smooth concrete surface, having a friction coefficient of $\mu = 0.5$ with the tile substrate.

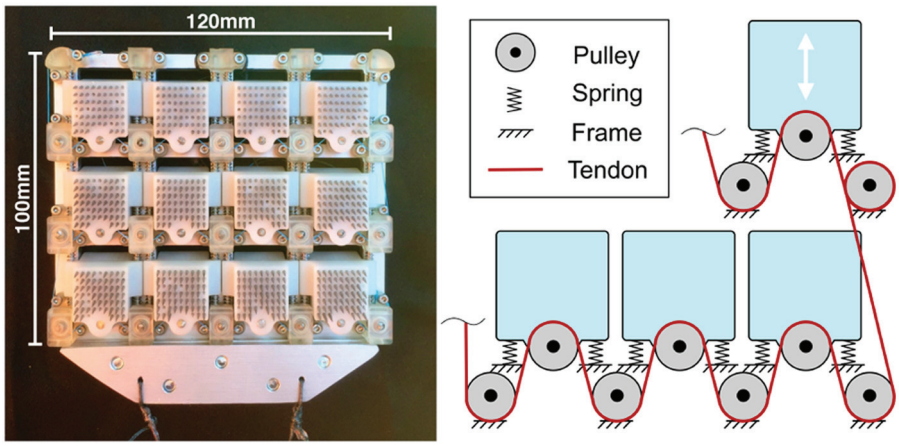


Fig. 16. Left: actual palm design. Right: diagram of the tendon-pulley system. The white arrow indicates the tile sliding direction.

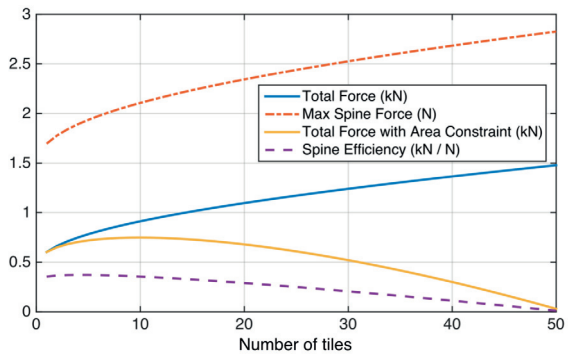


Fig. 17. Shear force (N) and spine efficiency (total palm force/maximal spine force of the entire palm) for 2000 spines with different numbers of tiles. *Area Constraint* and corresponding *Spine Efficiency* curves account for space (and spines on it) consumed by pulleys and sliders.

5.2. *Rock climbing palm*

The second application involves applying arrays of spines to the foot of a large legged robot. The challenge in this case is to support loads of hundreds of Newtons, and to do so in a mechanism sufficiently compact and light to be placed on the end of a robot leg. Because we are interested primarily in shear forces and because the robot can orient its leg as desired about an axis perpendicular to the surface, we use unidirectional spine arrays. However, as illustrated in Figure 7, simply making the arrays larger will not suffice due to the accumulating effects of backlash and imperfect load sharing.

s An alternative solution is to distribute the load among multiple "floating" spine tiles (Figure 16). Each tile can travel slightly to engage asperities up to few millimeters away and compensate for any backlash. Each tile slides up to 5mm on two short rails under the influence of a tendon that loads all the tiles in parallel. Compression springs return the tiles to their rest position when not loaded. The tendon runs over low-friction pulleys to create a differential mechanism that evenly distributes the load. When any tile slips, the travel needed for it to reengage results in a small motion of all the remaining tiles with respect to the frame. The pulleys and the tendon are located very close to the surface to minimize moments on the sliding bearings, which could inhibit tile motion.

To quantify the effect of tile configuration, the performances of 2000 spines divided into different numbers of tiles are computed based on the palm design and the spine model, with the assumption of perfect load sharing (Figure 17). We define the *spine efficiency* as the ratio of total palm force to maximal spine force of the entire palm. It is apparent that dividing into more tiles can achieve greater adhesion, but this benefit diminishes as the tile number grows. Recall also from Figure 7 that tile adhesion scales almost linearly when the number of spines is less than 40. Hence it is not worth using smaller tiles given that each tile consumes some space for its rails, pulley, etc. Given the stochasticity of spine adhesion, another benefit of this load sharing system is allowing spine tiles that are engaged with weak asperities to break their contacts and travel to catch better asperities. This feature increases the reliability of the palm.

With the design shown in Figure 16, up to 70 kg shear force on a rough concrete surface has been demonstrated with a 100 mm x 120 mm palm surface area (12 tiles).

6. Conclusion

We have presented a new spine mechanism that consists of dense of arrays of spines that travel freely perpendicular to a surface but are constrained in the lateral direction. In comparison to previous spine mechanisms they provide higher levels of shear stress (up to 298 kPa) and conform better to rough surfaces. We describe the geometric and physical parameters that determine whether a spine will engage an asperity and how the maximum force varies for a range of loading angles and surface profiles. Depending on the application, including the roughness of the surface, one can vary the spine diameter, included tip angle, inclination angle, etc. to increase the probability of engaging asperities over a range of loading angles.

However, spine/asperity interaction is inherently stochastic. Therefore, the expected force from a group of spines is computed from empirical distributions of spine forces taken at a particular loading angle and extrapolated to a range of angles. The predicted results match those from experiments, with significant variability based on the random distributions of asperities and asperity strengths.

Spines sustain loads primarily in shear. To sustain loads from any direction, including perpendicular to a surface, a solution is to use opposed arrays of linearly-constrained spines with an internal preload. Depending on the preload, and positions of stops that limit spine array travel, a three-dimensional limit surface in force space can be constructed for the mechanism. Any force vector within this limit surface will be sustained. However, the limit surface, like the two-dimensional spine loading results, is stochastic. When comparing it to empirical results for gripping a concrete surface with opposed arrays of spines the predicted values match the average values, but there is considerable trial-to-trial variation. A consequence of this variability is that when spines are used for critical applications like climbing they should be used in combination with force sensing; grips should be tested before trusting them to support the weight of a robot. The same protocol was necessary when climbing with earlier compliant spine arrays (e.g. for RiSE (Spenko et al., 2008)).

Two special mechanisms are presented for human-scale robotic locomotion. The first is a lightweight mechanism that places four arrays of spines at the tip of a pole to be used by a humanoid robot in rough terrain. The mechanism allows the spines to grip convex features as well as flat surfaces, providing an increased effective coefficient of friction as compared to a simple pole with a hard or rubber tip.

The second mechanism is an array of 12 floating unidirectional tiles for the palm or foot of a large legged robot. It provides high shear forces (up to 70 kg on concrete) in a compact (100 mm x 120 mm) package. The palm mechanism also illustrates a tradeoff in spine array design. When spine arrays grow too large, load sharing and the achievable maximum shear stress suffer. It works better to use moderate numbers of spines (up to 60 in the present case) for the arrays and to connect the arrays with a differential that divides the overall force evenly among them. Similar principles have been used when scaling gecko-inspired adhesives to human-scale loads (Hawkes et al., 2015).

For the climbing robot, the next step is to augment the spiney palm with fingers, modeled approximately after the SupraPed mechanism depicted in Figure 1 and Figure 14. With this addition, RoboSimian should be able to grasp convex "sloper" features in a manner similar to that used by human rock climbers (Luebben, 2004).

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Note

1. This is in contrast to a perching micro air vehicle that hangs from a gripper with spines (Jiang et al., 2014).

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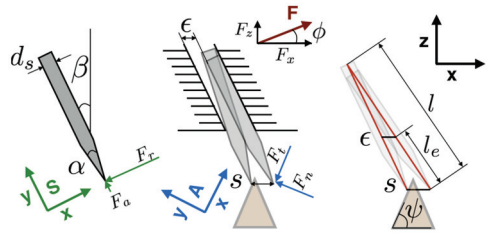


Fig. 18. Parameter Definitions: tip included angle α, spine shaft diameter d_s, spine inclination angle β, sliding channel tolerance ϵ, tip backlash s, spine length l, extended spine length l_e, contact surface slope ψ, loading force F, loading angle ϕ, loading force components F_x and F_z in the global coordinates, radial force F_r and axial force F_a in the spine coordinates S, and tangential force F_t and normal force F_n in the asperity coordinates A.

Appendices

A. Spine model definition

The main geometric parameters are illustrated in Figure 18 and include (i) spine tip included angle α; (ii) spine shaft diameter d_s; (iii) inclination angle of the spine β; (iv) spine stiffness with respect to the tip displacement k_t, (v) normal spring stiffness k_n; (vi) sliding channel tolerance ϵ; (vii) range of normal travel l_{e_max} and (viii) coefficient of friction between sliding channel and the spine μ_s.

In the model, we use ψ to denote the local slope of the contact surface at the spine tip and ϕ to designate the force application angle. Three different coordinate systems (Figure 18) are used in the models: (i) global coordinates where F_x and F_z are the shear and normal adhesion components of the spine; (ii) spine coordinates where F_r and F_a are the radius and axial force components at the tip, mainly for spine strength analysis, and (iii) asperity coordinates where F_t and F_n are the tangential and normal ground reaction forces with respect to the local surface.

B. Spine strength derivations

There are two locations on the spine subject to bending failure (Figure 3, highlighted in red). If the moment of inertia of the cross-section at the tip does not grow as rapidly as the bending moment, the bending stress $\sigma(l_e, d_s)$ reaches a local maximum at the upper contact where the spine exits the channel. From elementary beam theory (Crandall, 2012)

$$\sigma(l_e) = \frac{F_r l_e d_s/2}{\pi d_s^4/4} \tag{16}$$

where the loading force is limited by the extension length l_e and the shaft diameter.

The second local maximum can occur near the tip if $d(y)$ has not grown large enough. If the blunted tip diameter is d_m, the bending stress over the tapered portion is

$$\sigma(y) = \frac{F_r y}{2\pi(d_m/2 + \tan(\alpha/2)y)^3} \tag{17}$$

Based on its derivative, the local maximum occurs when $y = d_m/4\tan(\alpha/2)$ which should not exceed $\sigma_{(max)}$

$$\sigma\left(\frac{d_m}{4\tan(\alpha/2)}\right) = \frac{8F_r}{27\pi\tan(\alpha/2)\,d_m^2} < \sigma_{max} \tag{18}$$

which gives a lower bound to the tip included angle α.

C. Spine-channel interaction constraints

In the first constraint, the friction to hold the spine under a given loading force, without allowing the spine to slip along the channel needs to stay within the friction cone

$$\beta - \arctan\mu_s \le \phi \le \beta + \arctan\mu_s \tag{19}$$

where ϕ is the loading angle and μ_s is the coefficient of friction between sliding channel and spine. Note that the lower limit can become negative, in which case the channel friction on the spine points towards the surface. When the loading angle is too steep or too shallow, the spine starts to slide downward or upward with respect to the channel, in which case the adhesion created by the spine tip is constrained along the friction cone edge, which is the upper or lower limit of equation (19). The adhesion magnitude is reduced from F to F' by factor $C_c(\phi,\beta)$ which becomes 1 when there is no channel slip

$$F' = C_c(\phi,\beta)F = \sin(\beta + \frac{\pi}{2} - \phi - \arctan\mu_s)(1+\mu_s^2)F \tag{20}$$

For the second constraint, concerning trapped particles in the channel, there is a force balance, which is shown in Figure 19. If N_1 and N_2 represent normal forces applied on the particle by the spine surface and channel wall respectively, μ_1 and μ_2 are the corresponding coefficients of friction, and f_1 is friction force applied by the spine, the force balance becomes

$$\begin{cases} f_1\cos\frac{\alpha}{2} + N_1\sin\frac{\alpha}{2} = N_2\mu_2 \\ N_1\cos\frac{\alpha}{2} - f_1\sin\frac{\alpha}{2} = N_2 \end{cases} \tag{21}$$

The friction force from the spine should be no larger than its friction limit

$$f_1 \le N_1\mu_1 \tag{22}$$

Based on the above equations, a requirement on the coefficient of friction between the dust particle and spine can be derived

$$\mu_1 \ge \frac{\mu_2\cos\frac{\alpha}{2} - \sin\frac{\alpha}{2}}{\mu_2\sin\frac{\alpha}{2} + \cos\frac{\alpha}{2}} \equiv \mu_m(\mu_2,\alpha) \tag{23}$$

Solving the derivatives to find the relations between μ_1, μ_2 and α we have

$$\begin{cases} \frac{d\mu_m(\mu_2,\alpha)}{d\mu_2} = \frac{1}{(\mu_2\sin\frac{\alpha}{2}+\cos\frac{\alpha}{2})^2} > 0 \\ \frac{d\mu_m(\mu_2,\alpha)}{d\alpha} = -\frac{1+\mu_2^2}{2(\mu_2\sin\frac{\alpha}{2}+\cos\frac{\alpha}{2})^2} < 0 \end{cases} \tag{24}$$

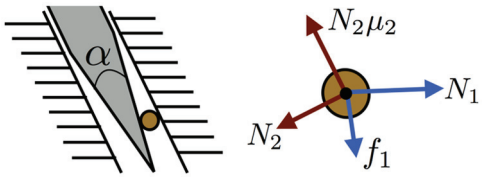


Fig. 19. Diagram and forces for dirt particles in the spine mechanism. Red arrows represent forces applied by the channel wall on the dust particle; blue arrows are applied by the spine surface.

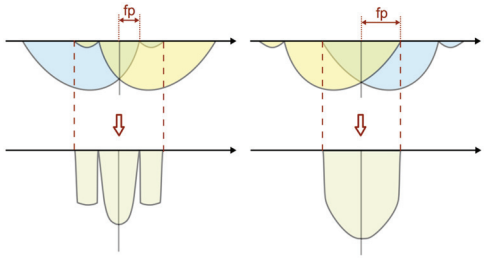


Fig. 20. The overall adhesion in the (x,z) direction can be computed by summing two mirrored single tile limit curves offset by $2\times$ the preload force. However, when either curve is zero, the result is zero.

D. Opposed gripper model derivations

Due to the opposed configuration, the individual limit surfaces of the two tiles are mirrored about the (y,z) plane. Because the gripper is rotationally constrained when being loaded, the total maximum normal adhesion is the sum of the two limit surfaces for forces in the z direction. However, for shear loading there is a spring to control the force distribution and the maximum adhesion needs to be modeled differently. If one tile fails, or produces zero force, the mechanism loses its grip. Thus the limit curve in the (x,z) plane is computed as shown in Figure 20, where we take the limit curve from each tile, invert one of them, and offset it by two times the preload force, f_p. If f_p becomes either too large or too small, the overall limit curve will be reduced as the force at one of the tiles becomes zero. Even with an appropriate preload, there are small angular regions in which an external load force can produce failure as one of the tiles loses its grip. Recall, however, that the spine/asperity contact and adhesion are probabilistic so that, in practice, these gaps may not be evident (e.g. as in Figure 12).

If the limit surface of each tile is represented by $f_z = F(f_x,f_y)$, the mechanism surface limit becomes

$$f_z = \begin{cases} F(f_{x1},\frac{f_y}{2}) + F(f_{x2},\frac{f_y}{2}) & \text{if } F(f_{x1},\frac{f_y}{2}) \geq 0 \\ & \text{and } F(f_{x2},\frac{f_y}{2}) \geq 0 \\ 0 & \text{else} \end{cases} \tag{25}$$

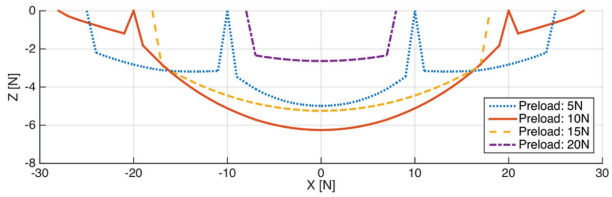


Fig. 21. 2D limit curve (x-z plane) of the opposed mechanism with various preload force levels (60 spines per tile, $\beta = 15^\circ$).

where f_{x1} and f_{x2} are given by

$$f_{x1} = f_p + f_x, \quad f_{x2} = f_p - f_x. \tag{26}$$

Note that only $f_x > 0$ is discussed here because the limit surface is mirrored about the (y,z) plane.

Figure 21 shows the effect of applying various amounts of internal preload. Increasing the preload from 5 N to 10 N allows a larger force in the z direction. However, increasing the preload further brings the tiles close to their individual limits, compromising the effectiveness of the system.

The above equations are valid only when none of the hard stops is touching a tile. When a tile encounters the outer hard stop, additional load forces will be borne entirely by that tile, hastening the point at which it (and therefore the entire mechanism) reaches its limit. Meanwhile, the force on the opposite tile is fixed, according to the spring length.

For example, when pulling in the positive x direction the left tile will reach its limit and the right tile will maintain a fixed force. Therefore the tile adhesion becomes bounded for equation (25)

$$f_{x2} = f_{hs2} \text{ if } f_{x1} \geq f_{hs1} \tag{27}$$

where f_{hs1} and f_{hs2} are the adhesion of the left and right tiles respectively when one of the tiles hits the outer hard stop. If inner hard stop is hit first, the additional force transmits to right tile instead. In such case, the bounding condition becomes

$$f_{x1} = f_{hs1} \text{ if } f_{x2} \leq f_{hs2} \tag{28}$$